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Tissue AI

A visual tool that automatically transforms your text into customizable illustrations

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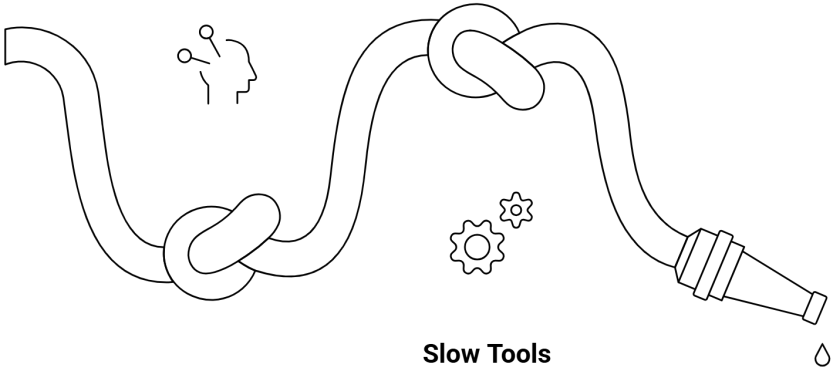
Problem Definition

Manual Creation

Time-consuming and mentally taxing



Slow Tools

Planning hinders clear thinking

→ Digital information grows very fast.

→ Users struggle to read, organize, and understand large text content.

→ Lecture notes, ideas, documents, and workflows become hard to analyze using only text.

→ Visual forms like mind maps and flowcharts improve understanding.

But the issue is

Objective

The goal is to design and develop Tissue AI.

Tissue AI is an intelligent system that converts text into visual diagrams. The system creates flowcharts and mind maps from written content.

The system reduces manual effort during diagram creation. It improves knowledge visualization using Gen AI techniques.

Methodology

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follows a **Generative AI-driven text-to-visualization pipeline.**

transforms unstructured natural language input into structured visual representations

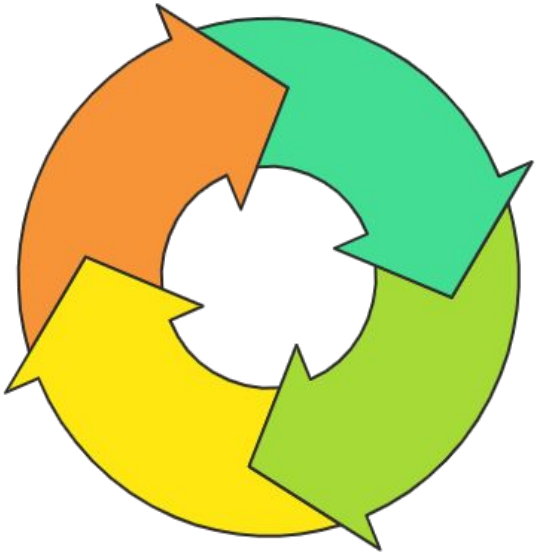
- **Methodological Approach**
  - Generative AI
  - NLP Techniques
  - Structured Intermediate Representation
- **Tissue AI**
- **Data Analysis Procedures**
  - Text Preprocessing
  - Semantic Understanding
  - Structure Generation
  - Tools and Technologies
- **Data Collection Techniques**
  - User-Provided Text
  - On-Demand Collection

does not rely on traditional datasets such as surveys or labeled training samples. Instead, it operates on **dynamic user-provided textual input.**

Project execution & QA

1. System design and GenAI model selection
2. Text preprocessing and prompt structuring
3. Generative relationship and structure generation
4. Diagram rendering using visualization frameworks
5. Testing and iterative refinement

Quality Assurance Cycle



Test Consistency

Ensure consistency across input types

Validate Logic

Ensure diagram logic is sound

Gather Feedback

Collect user feedback for refinement

Check Clarity

Verify visual clarity and readability

INNOVATION

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POINT 01

Generative AI to automatically convert unstructured text into structured visual representations

POINT 02

The system eliminates user-defined layouts by dynamically generating flowcharts and mind maps through AI-driven semantic understanding.

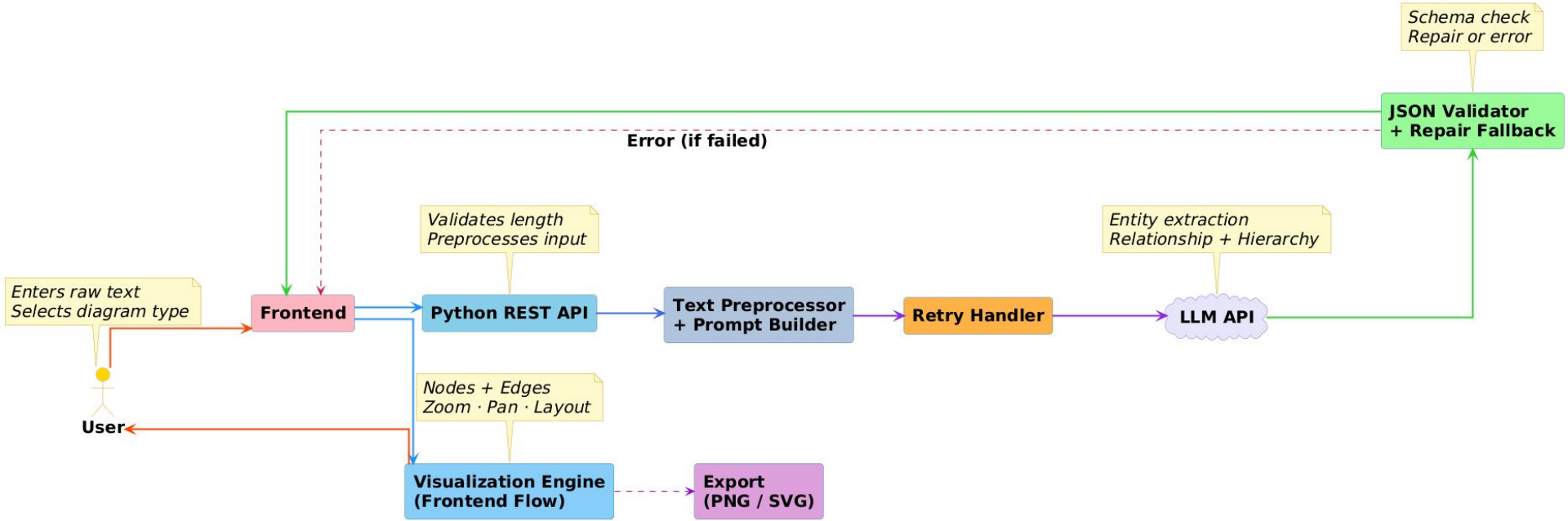
POINT 03

This enables faster learning, scalable understanding, and forms a foundation for future AI-assisted learning and visualization systems.

Software Requirements

Development	Large Language Models	Testing and Deployment
Python	OpenAI API	PyTest
JavaScript	Google Gemini API	Postman
React.js	Anthropic Claude API	Docker
Visual Studio Code	Open-source LLM APIs	GitHub Actions
Git		Vercel
GitHub		

System Architecture



Product Backlog

Feature	User Story	Priority
Text Input	User can enter multi-line text and trigger diagram generation	High
Preprocessing	System normalizes text, validates token size, and builds prompt	High
LLM Engine	System sends structured prompt to LLM and receives JSON output	High
JSON Validation	System validates schema and repairs malformed LLM response	Medium
Diagram Render	System renders flowchart and mind map with zoom and pan	High
Export	User can export rendered diagram as PNG or SVG file	Low

Software Requirements Specification - 1

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Overview

Single-user web app that turns unstructured text into flowcharts or mind maps using an LLM.

No database · No auth · Single user

System Components

- React Frontend
- Python REST API
- LLM API Integration
- Visualization Library

Functional Requirements

User Input

- Multi-line text input
- Character limit check
- Generate button
- Loading indicator

Preprocessing

- Whitespace cleanup
- Token size validation
- Prompt templating
- Retry logic

LLM Engine

- API integration
- JSON schema output
- Flowchart schema
- Mind map schema

Diagram Render

- Auto node placement
- Zoom & pan
- View switching
- Edge rendering

Software Requirements Specification - 2

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Out of Scope (MVP)

- ☐ User authentication
- ☐ Database storage
- ☐ Multi-user collaboration
- ☐ Offline LLM
- ☐ Enterprise infrastructure
- ☐ LMS integration

Deliverables

- ☐ Deployed web application
- ☐ Flowchart generation
- ☐ Mind map generation
- ☐ LLM processing pipeline
- ☐ Source code + documentation

LLM Structuring Engine

Flowchart Schema

nodes[] → id, label, type

edges[] → source, target

direction → LR or TB

Mind Map Schema

root → central topic

children[] → nested topics

level → depth indicator

Validation Pipeline

1. Parse JSON
2. Schema Check
3. Auto Repair
4. Re-validate
5. ☐ Render